regular and smooth than in the previous example because the sine wave is at a much higher frequency, resulting in fewer samples per cycle. Figure (b) shows the frequency spectrum of this signal. As you would expect, you can identify the fundamental and harmonics. This example shows that harmonics can extend to frequencies greater than 0.5 of the sampling frequency, and will be *aliased* to frequencies somewhere between 0 and 0.5. You don't notice them in (b) because their amplitudes are too low. Figure (c) shows the frequency spectrum plotted on a logarithmic scale to reveal these low amplitude aliased peaks. At first glance, this spectrum looks like random noise. It isn't; this is a result of the many harmonics overlapping as they are aliased.

It is important to understand that this example involves distorting a signal *after* it has been digitally represented. If this distortion occurred in an analog signal, you would remove the offending harmonics with an antialias filter *before* digitization. Harmonic aliasing is only a problem when nonlinear operations are performed directly on a discrete signal. Even then, the amplitude of these aliased harmonics is often low enough that they can be ignored.

The concept of harmonics is also useful for another reason: it explains why the DFT views the time and frequency domains as *periodic*. In the frequency domain, an N point DFT consists of $N/2+1$ equally spaced frequencies. You can view the frequencies *between* these samples as (1) having a value of zero, or (2) not existing. Either way they don't contribute to the synthesis of the time domain signal. In other words, a *discrete* frequency spectrum consists of *harmonics*, rather than a continuous range of frequencies. This requires the time domain to be periodic with a frequency equal to the lowest sinusoid in the frequency domain, i.e., the fundamental frequency. Neglecting the DC value, the lowest frequency represented in the frequency domain makes one complete cycle every N samples, resulting in the time domain being periodic with a period of N. In other words, if one domain is *discrete*, the other domain must be *periodic*, and vice versa. This holds for all four members of the Fourier transform family. Since the DFT views both domains as discrete, it must also view both domains as periodic. The samples in each domain represent harmonics of the periodicity of the opposite domain.

Chirp Signals

Chirp signals are an ingenious way of handling a practical problem in echo location systems, such as radar and sonar. Figure 11-9 shows the frequency response of the chirp system. The magnitude has a constant value of one, while the phase is a parabola:

EQUATION 11-7
Phase of the chirp system.

$$Phase\ X[k] = \alpha k + \beta k^2$$